ASSIGNMENT-BANKING SYSTEM
TASK 2

1. Insert at least 10 sample records into each of the following tables.
• Customers

INSERT INTO Customers (customer_id, first_name, last_name, DOB, email, phone_number, address) VALUES
(1, 'Harry', 'Potter', '1990-07-31', 'harry.potter@gmail.com', '9876543210', '4 Privet Drive, Little Whinging, Surrey'),
(2, 'Ron', 'Weasley', '1990-03-01', 'ron.weasley@gmail.com', '9845678901', 'The Burrow, Ottery St. Catchpole, Devon'),
(3, 'Hermione', 'Granger', '1989-09-19', 'hermione.granger@gmail.com', '9786543210', 'Hampstead Garden Suburb, London'),
(4, 'Sirius', 'Black', '1959-11-03', 'sirius.black@gmail.com', '9765432109', '12 Grimmauld Place, London'),
(5, 'Remus', 'Lupin', '1960-03-10', 'remus.lupin@gmail.com', '9876504321', '13th Avenue Road,London'),
(6, 'Lily', 'Evans', '1960-01-30', 'lily.evans@gmail.com', '9876543098', 'Spinner’s End, Cokeworth'),
(7, 'Luna', 'Lovegood', '1981-02-13', 'luna.lovegood@gmail.com', '9785463210', 'Lovegood House, Ottery St. Catchpole'),
(8, 'Narcissa', 'Malfoy', '1955-06-01', 'narcissa.malfoy@gmail.com', '9898765432', 'Malfoy Manor, Wiltshire'),
(9, 'James', 'Potter', '1960-03-27', 'james.potter@gmail.com', '9876234510', 'Godric’s Hollow, West Country, England'),
(10, 'Tom', 'Riddle', '1926-12-31', 'tom.riddle@gmail.com', '9887766543', 'Wool’s Orphanage, London');

```
ysql> SELECT * FROM Customers;
+-------------+------------+-----------+------------+---------------------------+--------------+---------------------------------------+
| customer_id | first_name | last_name | DOB        | email                     | phone_number | address                               |
+-------------+------------+-----------+------------+---------------------------+--------------+---------------------------------------+
           1 | Harry      | Potter    | 1990-07-31 | harry.potter@gmail.com    | 9876543210   | 4 Privet Drive, Little Whinging, Surrey |
           2 | Ron        | Weasley   | 1990-03-01 | ron.weasley@gmail.com     | 9845678901   | The Burrow, Ottery St. Catchpole, Devon |
           3 | Hermione   | Granger   | 1989-09-19 | hermione.granger@gmail.com | 9786543210   | Hampstead Garden Suburb, London       |
           4 | Sirius     | Black     | 1959-11-03 | sirius.black@gmail.com    | 9765432109   | 12 Grimmauld Place, London            |
           5 | Remus      | Lupin     | 1960-03-10 | remus.lupin@gmail.com     | 9876504321   | 13th Avenue Road,London               |
           6 | Lily       | Evans     | 1960-01-30 | lily.evans@gmail.com      | 9876543098   | Spinner?s End, Cokeworth              |
           7 | Luna       | Lovegood  | 1981-02-13 | luna.lovegood@gmail.com   | 9785463210   | Lovegood House, Ottery St. Catchpole  |
           8 | Narcissa   | Malfoy    | 1955-06-01 | narcissa.malfoy@gmail.com | 9898765432   | Malfoy Manor, Wiltshire               |
           9 | James      | Potter    | 1960-03-27 | james.potter@gmail.com    | 9876234510   | Godric?s Hollow, West Country, England |
          10 | Tom        | Riddle    | 1926-12-31 | tom.riddle@gmail.com      | 9887766543   | Wool?s Orphanage, London              |
+-------------+------------+-----------+------------+---------------------------+--------------+---------------------------------------+
0 rows in set (0.00 sec)
```

• Accounts

```
mysql> INSERT INTO Accounts (account_id, customer_id, account_type, balance) VALUES
    -> (1, 1, 'savings', 1500.00),
    -> (2, 2, 'current', 3200.50),
    -> (3, 3, 'savings', 7800.75),
    -> (4, 4, 'current', 777700.00),
    -> (5, 5, 'savings', 5400.00),
    -> (6, 6, 'current', 9000.25),
    -> (7, 7, 'savings', 1200.00),
    -> (8, 8, 'current', 4300.00),
    -> (9, 9, 'savings', 10000.00),
    -> (10, 10, 'zero_balance', 0.00);
Query OK, 10 rows affected (0.08 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Accounts;
+------------+-------------+--------------+-----------+
| account_id | customer_id | account_type | balance   |
+------------+-------------+--------------+-----------+
|          1 |           1 | savings      |   1500.00 |
|          2 |           2 | current      |   3200.50 |
|          3 |           3 | savings      |   7800.75 |
|          4 |           4 | current      | 777700.00 |
|          5 |           5 | savings      |   5400.00 |
|          6 |           6 | current      |   9000.25 |
|          7 |           7 | savings      |   1200.00 |
|          8 |           8 | current      |   4300.00 |
|          9 |           9 | savings      |  10000.00 |
|         10 |          10 | zero_balance |      0.00 |
+------------+-------------+--------------+-----------+
10 rows in set (0.03 sec)
```

• Transactions

```
mysql> INSERT INTO Transactions (transaction_id, account_id, transaction_type, amount, transaction_date) VALUES
    -> (1, 1, 'deposit', 500.00, '2024-03-20 10:15:00'),
    -> (2, 2, 'withdrawal', 200.00, '2024-03-21 14:30:00'),
    -> (3, 3, 'deposit', 1000.00, '2024-03-22 09:45:00'),
    -> (4, 4, 'transfer', 1500.00, '2024-03-23 11:10:00'),
    -> (5, 5, 'withdrawal', 600.00, '2024-03-24 16:20:00'),
    -> (6, 6, 'deposit', 2500.00, '2024-03-25 12:00:00'),
    -> (7, 7, 'withdrawal', 400.00, '2024-03-26 17:40:00'),
    -> (8, 8, 'transfer', 1800.00, '2024-03-27 15:05:00'),
    -> (9, 9, 'deposit', 5000.00, '2024-03-28 08:30:00'),
    -> (10, 10, 'withdrawal', 300.00, '2024-03-29 19:55:00');
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Transactions;
+----------------+------------+------------------+---------+---------------------+
| transaction_id | account_id | transaction_type | amount  | transaction_date    |
+----------------+------------+------------------+---------+---------------------+
|              1 |          1 | deposit          |  500.00 | 2024-03-20 10:15:00 |
|              2 |          2 | withdrawal       |  200.00 | 2024-03-21 14:30:00 |
|              3 |          3 | deposit          | 1000.00 | 2024-03-22 09:45:00 |
|              4 |          4 | transfer         | 1500.00 | 2024-03-23 11:10:00 |
|              5 |          5 | withdrawal       |  600.00 | 2024-03-24 16:20:00 |
|              6 |          6 | deposit          | 2500.00 | 2024-03-25 12:00:00 |
|              7 |          7 | withdrawal       |  400.00 | 2024-03-26 17:40:00 |
|              8 |          8 | transfer         | 1800.00 | 2024-03-27 15:05:00 |
|              9 |          9 | deposit          | 5000.00 | 2024-03-28 08:30:00 |
|             10 |         10 | withdrawal       |  300.00 | 2024-03-29 19:55:00 |
+----------------+------------+------------------+---------+---------------------+
10 rows in set (0.01 sec)
```

2. Write SQL queries for the following tasks:

1. Write a SQL query to retrieve the name, account type and email of all customers.

```
mysql> SELECT C.first_name, C.last_name, A.account_type, C.email
    -> FROM Customers C
    -> JOIN Accounts A ON C.customer_id = A.customer_id;
+------------+-----------+--------------+---------------------------+
| first_name | last_name | account_type | email                     |
+------------+-----------+--------------+---------------------------+
| Harry      | Potter    | savings      | harry.potter@gmail.com    |
| Ron        | Weasley   | current      | ron.weasley@gmail.com     |
| Hermione   | Granger   | savings      | hermione.granger@gmail.com |
| Sirius     | Black     | current      | sirius.black@gmail.com    |
| Remus      | Lupin     | savings      | remus.lupin@gmail.com     |
| Lily       | Evans     | current      | lily.evans@gmail.com      |
| Luna       | Lovegood  | savings      | luna.lovegood@gmail.com   |
| Narcissa   | Malfoy    | current      | narcissa.malfoy@gmail.com |
| James      | Potter    | savings      | james.potter@gmail.com    |
| Tom        | Riddle    | zero_balance | tom.riddle@gmail.com      |
+------------+-----------+--------------+---------------------------+
10 rows in set (0.01 sec)
```

2. Write a SQL query to list all transaction corresponding customer.

```
mysql> SELECT C.first_name, C.last_name, T.transaction_type, T.amount, T.transaction_date
    -> FROM Customers C
    -> JOIN Accounts A ON C.customer_id = A.customer_id
    -> JOIN Transactions T ON A.account_id = T.account_id;
+------------+-----------+------------------+---------+---------------------+
| first_name | last_name | transaction_type | amount  | transaction_date    |
+------------+-----------+------------------+---------+---------------------+
| Harry      | Potter    | deposit          |  500.00 | 2024-03-20 10:15:00 |
| Ron        | Weasley   | withdrawal       |  200.00 | 2024-03-21 14:30:00 |
| Hermione   | Granger   | deposit          | 1000.00 | 2024-03-22 09:45:00 |
| Sirius     | Black     | transfer         | 1500.00 | 2024-03-23 11:10:00 |
| Remus      | Lupin     | withdrawal       |  600.00 | 2024-03-24 16:20:00 |
| Lily       | Evans     | deposit          | 2500.00 | 2024-03-25 12:00:00 |
| Luna       | Lovegood  | withdrawal       |  400.00 | 2024-03-26 17:40:00 |
| Narcissa   | Malfoy    | transfer         | 1800.00 | 2024-03-27 15:05:00 |
| James      | Potter    | deposit          | 5000.00 | 2024-03-28 08:30:00 |
| Tom        | Riddle    | withdrawal       |  300.00 | 2024-03-29 19:55:00 |
+------------+-----------+------------------+---------+---------------------+
10 rows in set (0.00 sec)
```

3. Write a SQL query to increase the balance of a specific account by a certain amount

```
ysql> UPDATE Accounts SET balance = balance + 500  WHERE account_id = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

ysql> SELECT * FROM ACCOUNTS;
+------------+-------------+--------------+-----------+
| account_id | customer_id | account_type | balance   |
+------------+-------------+--------------+-----------+
|          1 |           1 | savings      |   2000.00 |
|          2 |           2 | current      |   3200.50 |
|          3 |           3 | savings      |   7800.75 |
|          4 |           4 | current      | 777700.00 |
|          5 |           5 | savings      |   5400.00 |
|          6 |           6 | current      |   9000.25 |
|          7 |           7 | savings      |   1200.00 |
|          8 |           8 | current      |   4300.00 |
|          9 |           9 | savings      |  10000.00 |
|         10 |          10 | zero_balance |      0.00 |
+------------+-------------+--------------+-----------+
10 rows in set (0.00 sec)
```

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4. Write a SQL query to Combine first and last names of customers as a full_name.

```
mysql> SELECT CONCAT(first_name, ' ', last_name) AS full_name
    -> FROM Customers;
+------------------+
| full_name        |
+------------------+
| Harry Potter     |
| Ron Weasley      |
| Hermione Granger |
| Sirius Black     |
| Remus Lupin      |
| Lily Evans       |
| Luna Lovegood    |
| Narcissa Malfoy  |
| James Potter     |
| Tom Riddle       |
+------------------+
10 rows in set (0.01 sec)
```

5. **Write a SQL query to remove accounts with a balance of zero where the account type is savings.**(no records deleted in table)

```
mysql> DELETE FROM Accounts
    -> WHERE balance = 0 AND account_type = 'savings';
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM ACCOUNTS;
+------------+-------------+--------------+-----------+
| account_id | customer_id | account_type | balance   |
+------------+-------------+--------------+-----------+
|          1 |           1 | savings      |   2000.00 |
|          2 |           2 | current      |   3200.50 |
|          3 |           3 | savings      |   7800.75 |
|          4 |           4 | current      | 777700.00 |
|          5 |           5 | savings      |   5400.00 |
|          6 |           6 | current      |   9000.25 |
|          7 |           7 | savings      |   1200.00 |
|          8 |           8 | current      |   4300.00 |
|          9 |           9 | savings      |  10000.00 |
|         10 |          10 | zero_balance |      0.00 |
+------------+-------------+--------------+-----------+
```

6. Write a SQL query to Find customers living in a specific city.

```
mysql> SELECT *
    -> FROM Customers
    -> WHERE address LIKE '%London%';
+-------------+------------+-----------+------------+---------------------------+--------------+--------------------------------+
| customer_id | first_name | last_name | DOB        | email                     | phone_number | address                        |
+-------------+------------+-----------+------------+---------------------------+--------------+--------------------------------+
|           3 | Hermione   | Granger   | 1989-09-19 | hermione.granger@gmail.com | 9786543210   | Hampstead Garden Suburb, London |
|           4 | Sirius     | Black     | 1959-11-03 | sirius.black@gmail.com     | 9765432109   | 12 Grimmauld Place, London      |
|           5 | Remus      | Lupin     | 1960-03-10 | remus.lupin@gmail.com      | 9876504321   | 13th Avenue Road,London         |
|          10 | Tom        | Riddle    | 1926-12-31 | tom.riddle@gmail.com       | 9887766543   | Wool?s Orphanage, London        |
+-------------+------------+-----------+------------+---------------------------+--------------+--------------------------------+
4 rows in set (0.01 sec)
```

7. Write a SQL query to Get the account balance for a specific account.

```
mysql> SELECT balance
    -> FROM Accounts
    -> WHERE account_id = 1;
+---------+
| balance |
+---------+
| 2000.00 |
+---------+
1 row in set (0.01 sec)
```

8. Write a SQL query to List all current accounts with a balance greater than $1,000.

```
mysql> SELECT *
    -> FROM Accounts
    -> WHERE account_type = 'current' AND balance > 1000;
+------------+-------------+--------------+-----------+
| account_id | customer_id | account_type | balance   |
+------------+-------------+--------------+-----------+
|          2 |           2 | current      |   3200.50 |
|          4 |           4 | current      | 777700.00 |
|          6 |           6 | current      |   9000.25 |
|          8 |           8 | current      |   4300.00 |
+------------+-------------+--------------+-----------+
4 rows in set (0.00 sec)
```

9. Write a SQL query to Retrieve all transactions for a specific account

```
mysql> SELECT *
    -> FROM Transactions
    -> WHERE account_id = 1;
+----------------+------------+------------------+--------+---------------------+
| transaction_id | account_id | transaction_type | amount | transaction_date    |
+----------------+------------+------------------+--------+---------------------+
|              1 |          1 | deposit          | 500.00 | 2024-03-20 10:15:00 |
+----------------+------------+------------------+--------+---------------------+
1 row in set (0.01 sec)
```

10. Write a SQL query to Calculate the interest accrued on savings accounts based on a given interest rate.

```
mysql> SELECT account_id, balance, (balance * 0.05) AS interest_accrued  FROM Accounts WHERE account_type = 'savings';
+------------+----------+-----------------+
| account_id | balance  | interest_accrued |
+------------+----------+-----------------+
|          1 |  2000.00 |        100.0000 |
|          3 |  7800.75 |        390.0375 |
|          5 |  5400.00 |        270.0000 |
|          7 |  1200.00 |         60.0000 |
|          9 | 10000.00 |        500.0000 |
+------------+----------+-----------------+
5 rows in set (0.00 sec)
```

11. Write a SQL query to Identify accounts where the balance is less than a specified overdraft limit.

```
mysql> SELECT * FROM Accounts WHERE balance < 500;
+------------+-------------+--------------+---------+
| account_id | customer_id | account_type | balance |
+------------+-------------+--------------+---------+
|         10 |          10 | zero_balance |    0.00 |
+------------+-------------+--------------+---------+
1 row in set (0.01 sec)
```

12. Write a SQL query to Find customers not living in a specific city.

```
mysql> SELECT * FROM Customers WHERE address NOT LIKE '%London%';
+-------------+------------+-----------+------------+-------------------------+--------------+----------------------------------------+
| customer_id | first_name | last_name | DOB        | email                   | phone_number | address                                |
+-------------+------------+-----------+------------+-------------------------+--------------+----------------------------------------+
|           1 | Harry      | Potter    | 1990-07-31 | harry.potter@gmail.com  | 9876543210   | 4 Privet Drive, Little Whinging, Surrey |
|           2 | Ron        | Weasley   | 1990-03-01 | ron.weasley@gmail.com   | 9845678901   | The Burrow, Ottery St. Catchpole, Devon |
|           6 | Lily       | Evans     | 1960-01-30 | lily.evans@gmail.com    | 9876543098   | Spinner?s End, Cokeworth               |
|           7 | Luna       | Lovegood  | 1981-02-13 | luna.lovegood@gmail.com | 9785463210   | Lovegood House, Ottery St. Catchpole   |
|           8 | Narcissa   | Malfoy    | 1955-06-01 | narcissa.malfoy@gmail.com | 9898765432 | Malfoy Manor, Wiltshire                |
|           9 | James      | Potter    | 1960-03-27 | james.potter@gmail.com  | 9876234510   | Godric?s Hollow, West Country, England |
+-------------+------------+-----------+------------+-------------------------+--------------+----------------------------------------+
6 rows in set (0.01 sec)
```